BSAS 4th Semester

Assignment 1

The Profit Leak Report

SuperStore | 2023–2026

Analytical Problem Definition & Dashboard Submission

Individual Submission | April 2026

1. PROBLEM STATEMENTS

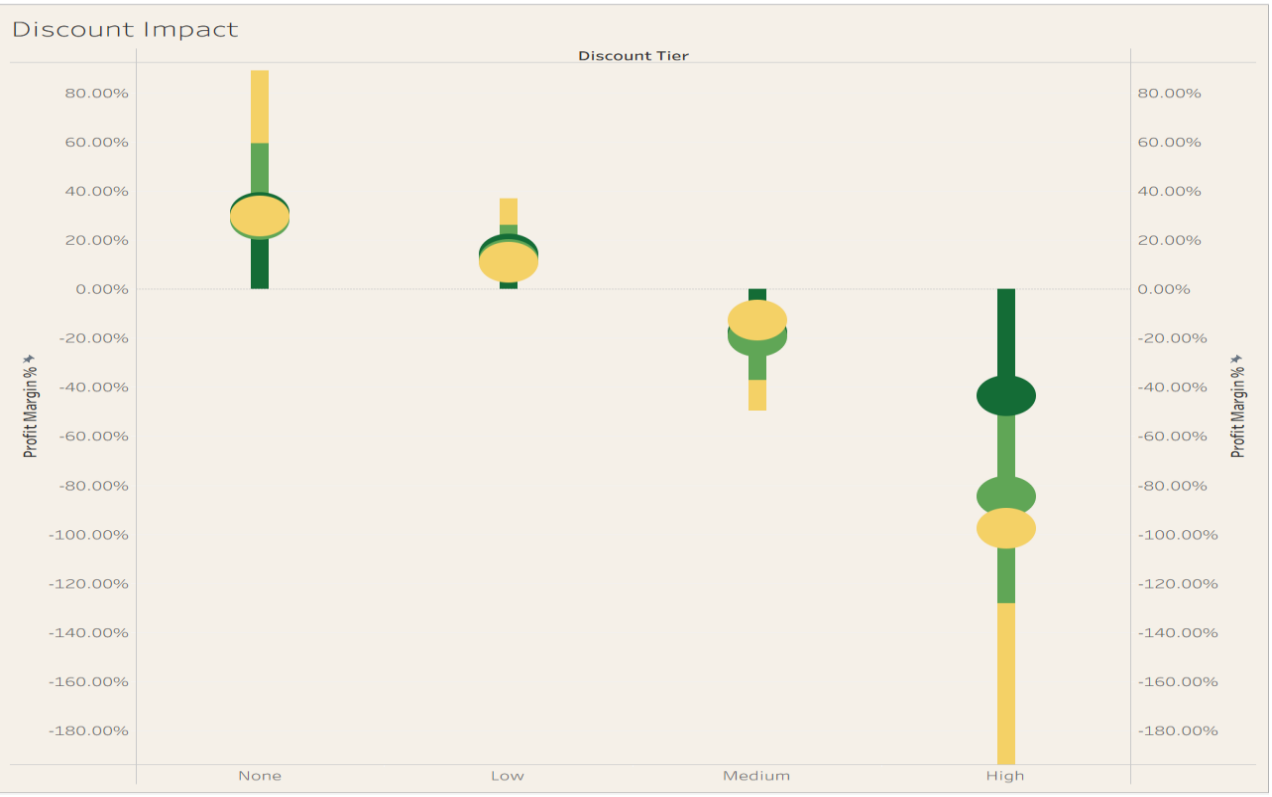
Before building any visualisation, five analytical questions were defined to guide the analysis. The goal was to ensure that every chart on the dashboard answers a specific business question rather than merely displays data. Each problem statement was designed to identify a potential area of profit loss. The five statements are presented below, along with the chart or visual used to answer them and the key calculated fields that support the analysis.

Problem statement 1 - How does discounting impact sales and profit, and does this vary across customer segments?"

Discounts are commonly offered under the assumption that lower prices lead to higher sales and more competition. Nevertheless, profit margins may decrease with discounts, and the increased sales scheme is not always successful. This problem statement investigates whether discounting in the SuperStore data is positively enhancing sales performance or primarily decreasing profit. It also verifies the differences in the impact of discounting across customer segments, as Consumer, Corporate, and Home Office customers might not respond to discounts the same way. If there are segments with poor sales growth and high profit losses, it indicates that the existing discounting policy is causing unnecessary profit erosion. The objective is to determine whether discounting can be constrained or adjusted by customer segment. Through this analysis, it is possible to determine which discounting instances are warranted and which constitute unnecessary revenue losses. It also helps to make better pricing decisions and more specific discount policies.

Visualisation used: A lollipop chart showing average Profit Margin % by Discount Tier (None, Low, Medium, High).

Calculated fields: Profit Margin %, Discount Tier



Problem Statement 2 — Which products are quietly losing money while appearing to contribute?

Revenue and profit are fundamentally different, yet in practice, they are often treated as proxies for one another. A product can appear healthy — generating consistent orders and contributing to the top-line — while actually running at a negative margin on every sale. This problem statement asks the data to distinguish between sub-categories that are genuinely value-generating and those that are silently destroying profit. The concern is that high-revenue, low-margin or negative-margin products may be receiving disproportionate sales effort and operational attention, while truly profitable products are underinvested.

Visualisation used: Treemap — sub-categories sized by total Sales, coloured by Profit Margin % (green for profitable, red for loss-making).

Calculated fields: Profit Margin %, Product Profit Status

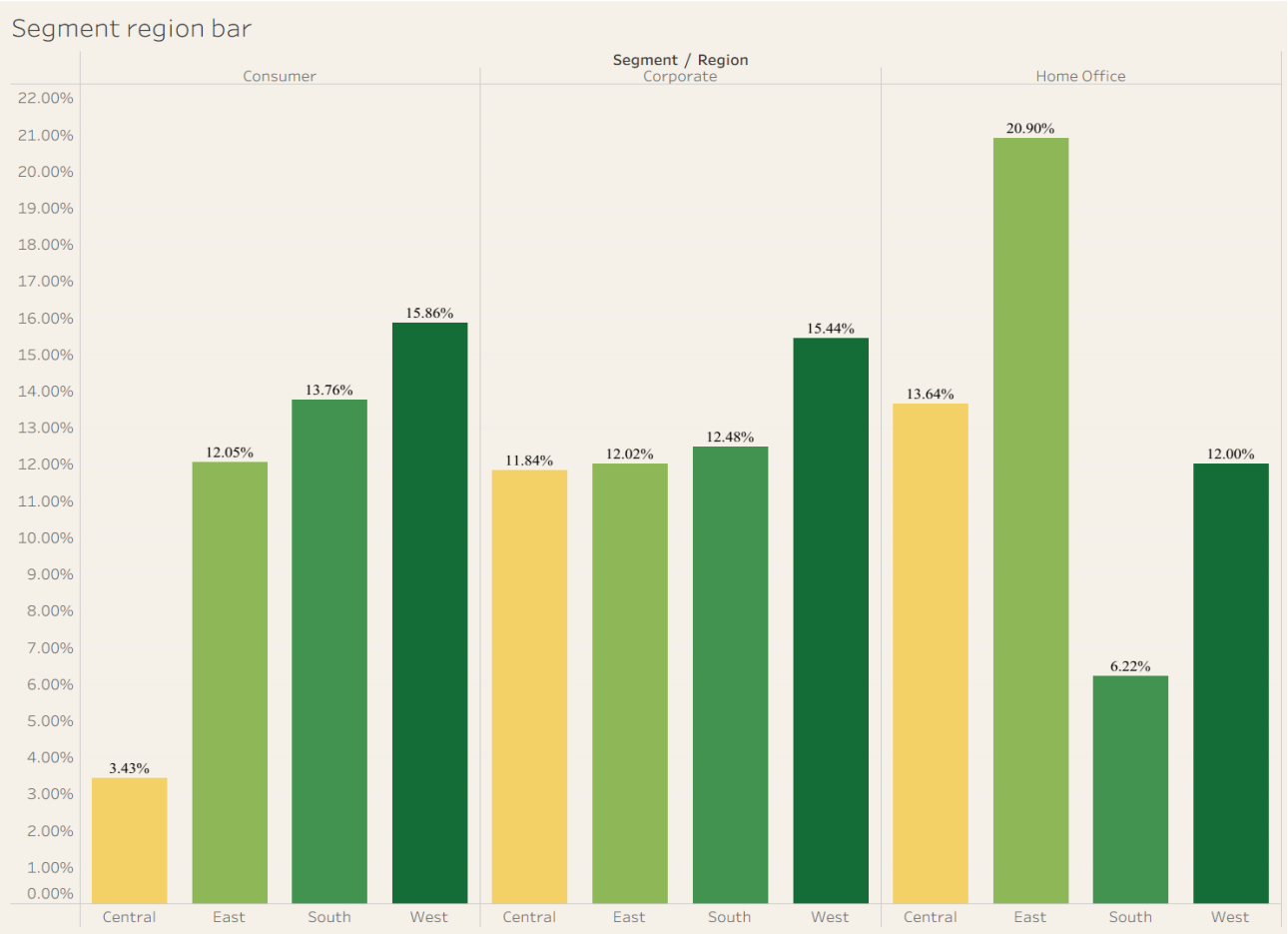


Problem Statement 3 — Do we lose money on certain customer types in certain regions?

Profitability at the aggregate level can mask significant variation at a more granular level. A customer segment that appears profitable when viewed nationally may in fact be loss-making in one or more specific regions — a pattern that would remain invisible in a top-line view. This problem statement investigates whether there are identifiable region-segment combinations where the business consistently operates at a loss. Such combinations represent a structural problem: the company is not just experiencing a temporary dip in performance, but repeatedly serving certain customers in certain geographies at a net cost.

Visualisation used: Grouped bar chart — four regions on the X-axis, profit margin % on the Y-axis, bars segmented by Consumer, Corporate, and Home Office. Bars coloured by profit status.

Calculated fields: Profit Margin %, Segment-Region Profit Flag

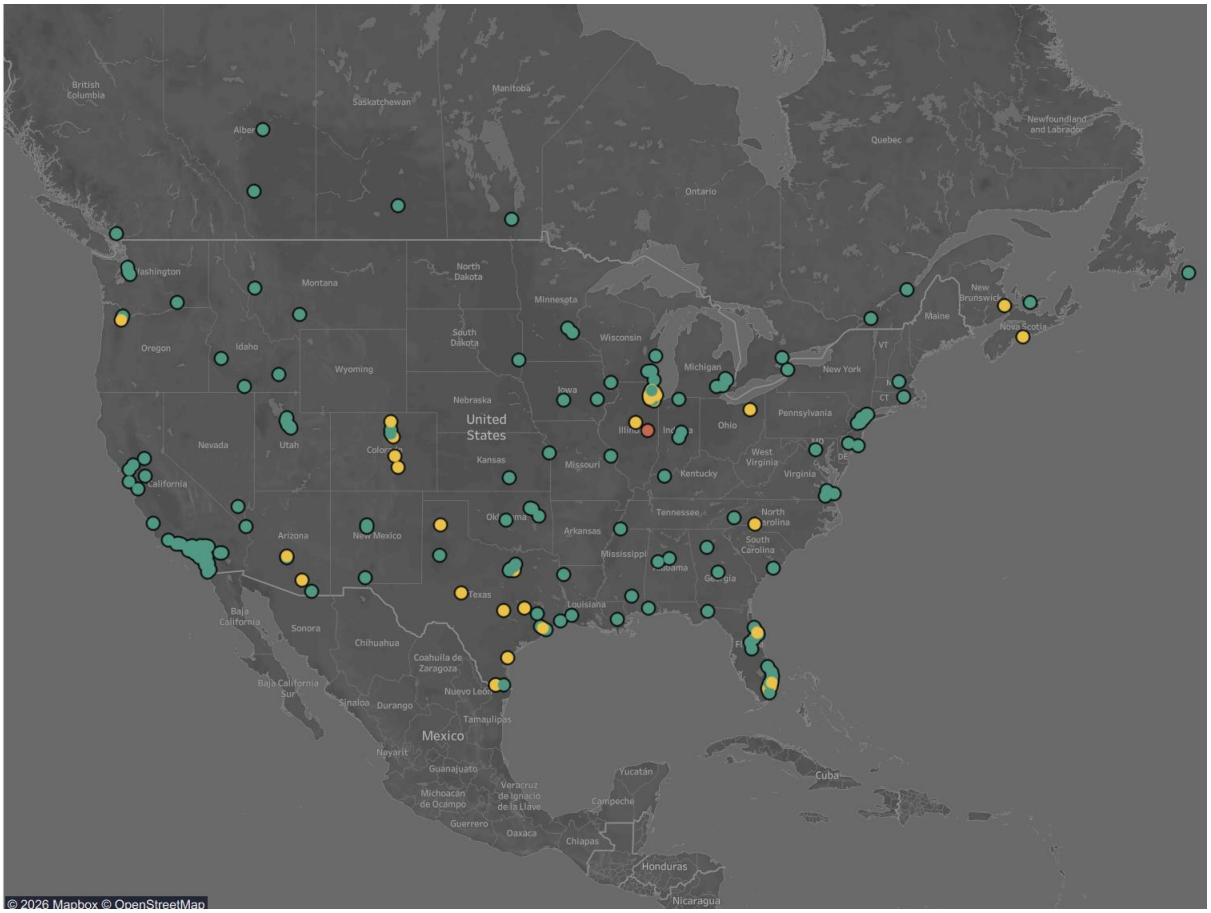


Problem Statement 4 — Are our busiest cities actually our most profitable, or just the most active?

A city generating large sales volumes naturally appears important. Sales teams are deployed there, promotions are run, and logistics costs are incurred — all on the premise that the city is a strong market. But high revenue does not guarantee high profit. When discounts, returns, and fulfilment costs are accounted for, some of the most active cities may actually be running at a loss. This problem statement distinguishes between cities where the business genuinely earns and those where it is merely busy. The practical implication is that resources — sales effort, promotional spend, inventory — may be misallocated toward high-volume but low-value markets.

Visualisation used: Geographic map — city-level bubbles, sized by total Sales, coloured by City Profit Status (green for profitable, red/amber for loss-making or marginal).

Calculated fields: City Profit Status, Profit Margin %



2. STRATEGY & APPROACH

2.1 Framing the Analysis

The analysis was structured around a single organising theme: profit leakage. This framing had a direct consequence for every subsequent decision — which charts to build, which calculated fields to create, and how the dashboard should be laid out. Every element needed to either locate a leak or explain its cause. The SuperStore data set consists of four years of transaction information on three customer groups, four geographical locations, and 17 product subclasses.

2.2 Chart Selection Logic

All the chart types were chosen because each was closely aligned with the corresponding problem statement. This was not a random limitation - the question was asked first, and the chart was selected to suit the question, not the other way around.

The plot for Problem Statement 1 was selected because correlation questions cannot be evaluated without both variables present simultaneously. A bar chart that summarised sales by discount level would have answered a slightly different question: it would have presented the average sales for each level. Nonetheless, that would have blotted out the individual-order-level association between discount and volume. A summary trend line complements the granularity retained in the lollipop chart.

The treemap of Problem Statement 3 was selected as it simultaneously encodes two dimensions: the size of each block is sales contribution, and the colour corresponds to the status of margin. This implies that a spectator can instantly spot the worst combination, a large red

block, without any more elaboration. The straightforward bar graph, ordered by the margin, would not have conveyed the relative sizes of the revenue for each sub-category.

The dumbbell chart applied in the Discount Impact panel has been adopted instead of a simple average bar chart because it shows the central tendency and spread of margins for each discount level. A simple average can mask drastic variation; the dumbbell format is more transparent about the distribution of results at each level. The grouped bar chart for Problem Statement 4 was chosen to preserve both analytical dimensions simultaneously — region and segment — without collapsing either into an aggregate. A stacked bar would have made it impossible to compare segment performance within a region. A separate chart for each segment would have required the viewer to perform cross-region comparisons mentally. The grouped format is the only layout that makes both comparisons directly visible.

2.3 Dashboard Construction

The dashboard was structured to support a logical reading path. The KPI strip at the top — showing Total Sales, Total Profit, Average Discount Given, and Loss Order Rate — is designed to establish context within the first five seconds of viewing. A reader should be able to understand the scale of the business and the problem before engaging with any individual chart.

The city map occupies the central panel of the dashboard. Its role is not purely analytical — it also serves as a spatial anchor, giving the viewer an immediate geographic frame of reference before they examine the more detailed charts below. The three bottom panels were arranged side by side to allow cross-chart comparison: a viewer can simultaneously see how profitability breaks down by customer segment, by product, and by discount behaviour.

Colour was treated as a data encoding, not a design choice. Green shades consistently indicate profitability; yellow indicates marginal or at-risk performance; red indicates a loss. This logic is maintained across the map, the treemap, and the segment bar charts, so that the viewer does not need to re-learn the colour meaning from chart to chart. Filter controls were added for Segment, Product Profit Status, and Discount Profit Status to allow investigation — the dashboard is intended to support analytical work, not just passive viewing.

3. CALCULATED FIELDS

All calculated fields were created to answer specific analytical questions. None were added as generic additions to the workbook. Each field powers at least one visualisation or filter in the dashboard, and several are reused across multiple charts. The complete list is set out in the table below.

Field Name	Formula / Definition	Purpose
Profit Margin %	`SUM([Profit]) / SUM([Sales])`	Core profitability metric used across all five charts and all four KPI cards.
Discount Tier	`IF [Discount] = 0 THEN "None" ELSEIF [Discount] <= 0.2 THEN "Low" ELSEIF [Discount] <= 0.4 THEN "Medium" ELSE "High" END`	Categorises continuous discount values into four actionable tiers for PS1 and PS2.
Product Profit Status	`IF SUM([Profit]) > 0 THEN "Profitable" ELSE "Loss-Making" END`	Binary classification is used to colour the treemap in PS3.
Segment Profit Flag	`IF [Profit Margin %] < 0 THEN "Loss" ELSEIF [Profit Margin %] < 0.05 THEN "At Risk" ELSE "Healthy" END`	Three-tier classification is used to identify danger zones in the PS4 segment-region analysis.
City Profit Status	`IF SUM([Profit]) > 0 THEN "Profitable" ELSE "Loss-Making" END`	Controls bubble colour on the city map for PS5.
Loss Order Rate	`COUNT(IF [Profit] < 0 THEN [Order ID] END) / COUNT([Order ID])`	KPI card — proportion of individual orders that lost money.
Avg Discount Given	`AVG([Discount])`	KPI card — provides overall context for the scale of discounting across the business.

4. KEY INSIGHTS

The dashboard surfaces five findings that are directly visible in the visualisations through labels, colour coding, and chart positions. Each insight below corresponds to one or more charts in the dashboard and is expressed as a finding with a practical implication.

Insight 1 — Discounting beyond 20% destroys margin with no volume payoff

The Discount Impact lollipop chart shows a definite elasticity, with profit margins declining as the discount level increases. There are usually no discount orders, and low-percentage discount orders tend to be profitable, with margins of at least zero. But after discounts enter the medium category, profitability declines considerably and reaches and/or even less than zero. The most discounted category is the least profitable, with consistently negative profit margins. This implies that heavy discounting is closely associated with loss transactions and

significant profit losses. The finding implies that high-tier discounting must be limited and applied only when the business has a strong reason to forego the margin.



Insight 2 — Tables and Supplies are the two largest product-level profit leaks

The treemap (PS3) shows Tables at -8.53% and Supplies at -2.51%. They both look like red blocks in the treemap. Their contribution to overall revenue is not extreme, and therefore, they do not stand out on a top-line graph, but are always and structurally unprofitable. Any Table Transaction or Supplies Sale decreases the profit. This is of interest, especially since these losses compare unfavourably with the good performances of Copiers (37.21) and Paper (43.39), which are doing well. The company is subsidising the products that are performing poorly with those that are performing well.



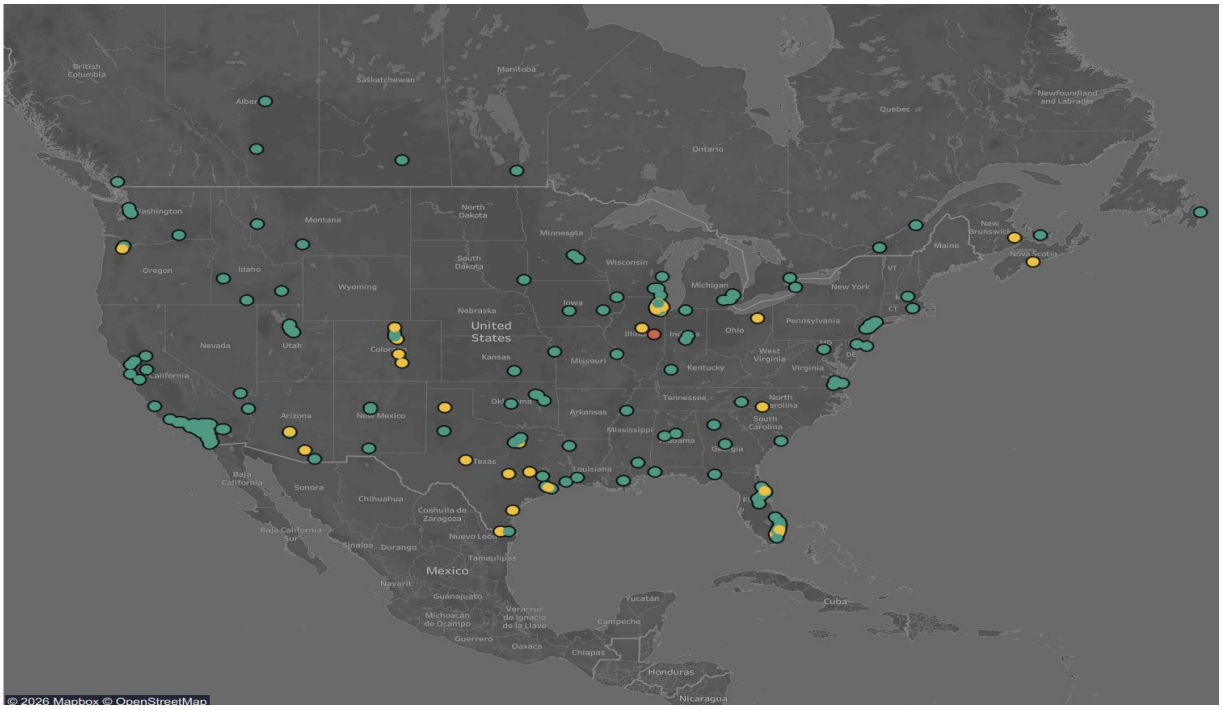
Insight 3 — Home Office in the South is a loss-making region-segment combination

The segment-region bar chart (PS4) indicates that the Home Office customer segment in the South region has a negative or near-zero profit margin. The Home Office segment itself is marginally profitable in other regions, indicating that the issue lies in the South region in particular, such as higher fulfilment costs, a different product mix, or more aggressive local discounting practices. This is ultimately what, in aggregate reporting, would remain in the dark, but becomes obvious when profitability is disaggregated across both dimensions simultaneously.



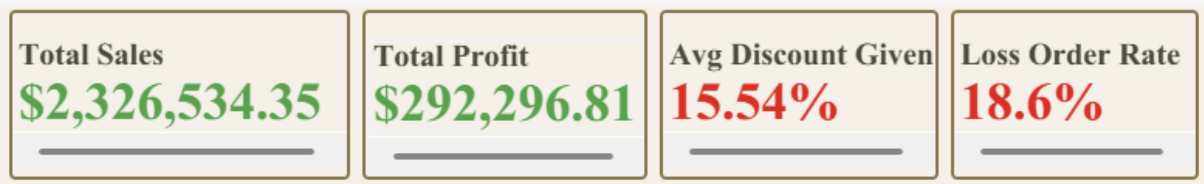
Insight 4 — Several high-revenue cities are running at a loss

The city map (PS5) shows a pattern that directly contradicts the assumption that large bubbles, such as high-sales cities, are the business's strongest markets. Several cities with substantial sales volumes are coloured red, indicating negative overall profit. These cities are generating significant operational activity like orders, fulfilment, customer service, but not profit. Allocating sales resources and promotional spend solely based on revenue would therefore direct investment toward the business's least profitable markets. The map provides a direct, city-by-city corrective to that misallocation.



Insight 5 — Nearly 1 in 5 orders lose money, which is not an anomaly.

The Loss Order Rate KPI, displayed at 18.6% in the top strip of the dashboard, is the single most important number in the report. At that level, approximately 1 in 5 transactions results in a loss rather than a profit. This is not a seasonal spike, but it reflects a structural pattern across the dataset. The rate is driven by the combination of excessively high-tier discounting and the sale of inherently low-margin products in regions and segments with disproportionately high costs. Addressing this number requires action on at least two fronts: a reformed discount policy and a review of the products actively sold and promoted.



5.2 Filters and Interactivity

Three filter controls are placed above the bottom panel. The Highlight Segment filter isolates a single customer type and allows the selection across all charts simultaneously. The Highlight Product Profit Status filter lets the viewer focus on loss-making or profitable products. The Highlight Discount Profit Status filter cross-filters by whether an order, in a given discount tier, resulted in profit or loss. All filters are cross-chart, meaning a selection in one panel updates the others. This makes the dashboard an interactive tool rather than a mere report.

Conclusion

The Profit Leak Report highlights that strong sales performance does not automatically translate into strong profitability. The dashboard findings show that profit leakage is mainly driven by aggressive discounting, loss-making product sub-categories, and specific region-segment combinations that consistently underperform. Several high-sales cities also generate low or negative profit, proving that business activity alone is not a reliable measure of success. Overall, the analysis suggests that SuperStore must shift focus from revenue growth to margin protection by tightening discount policies, reviewing unprofitable product lines, and targeting operational improvements in weak regions and customer segments. Addressing these areas will reduce loss-making orders and improve long-term profitability.